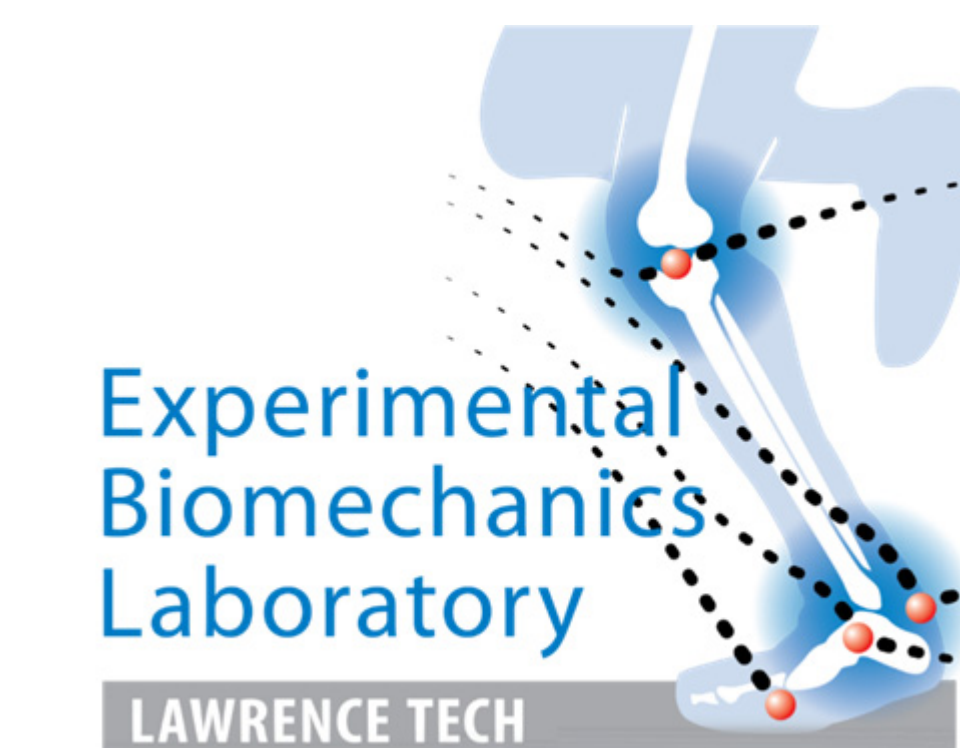




Finite Element Analysis Comparison of Two Types of Removable Partial Denture Designs

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Abstract

- Finite Element Analysis (FEA) was used to compare various removable partial dentures
- Comparison of reaction forces in abutment teeth

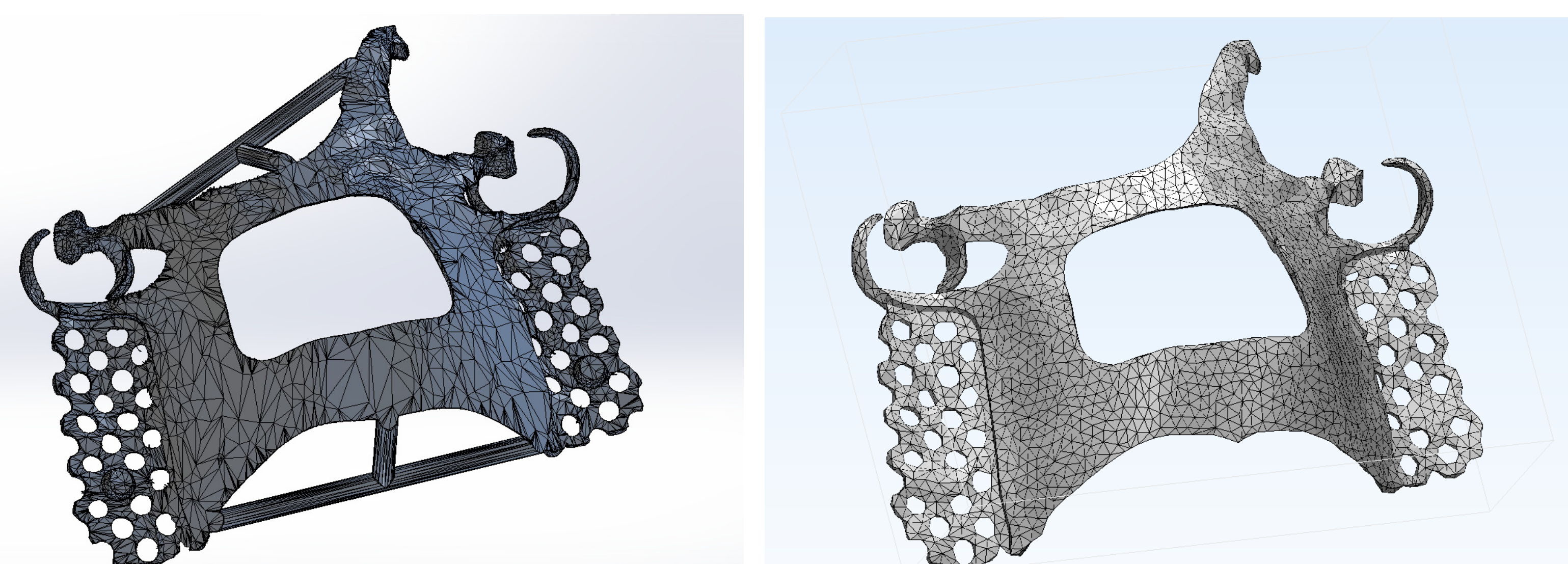
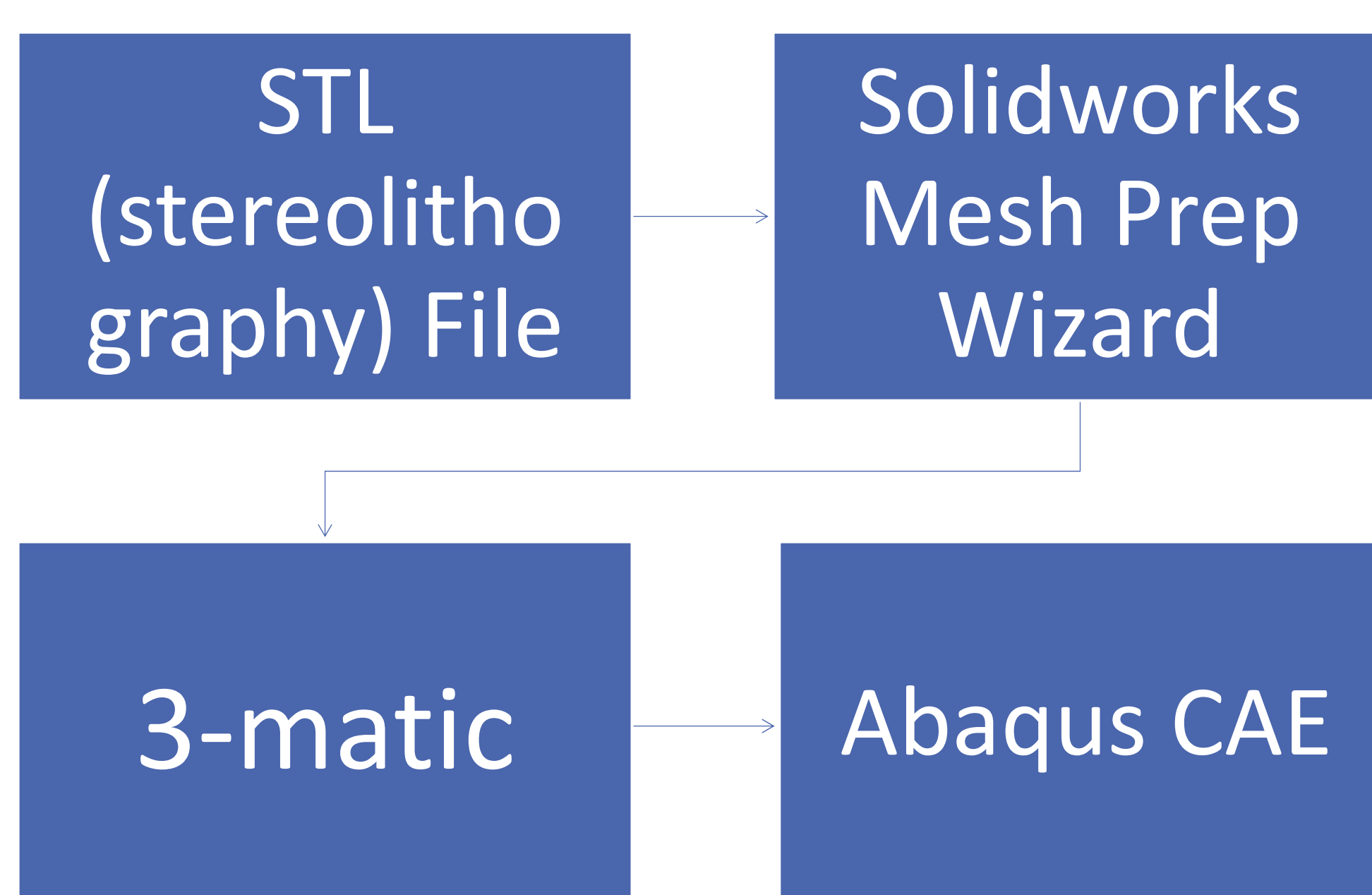
Background

- FEA can be used as initial comparison or as further evidence to back up other types of studies
- Cost effective and faster
- Conditions kept similar with each trial

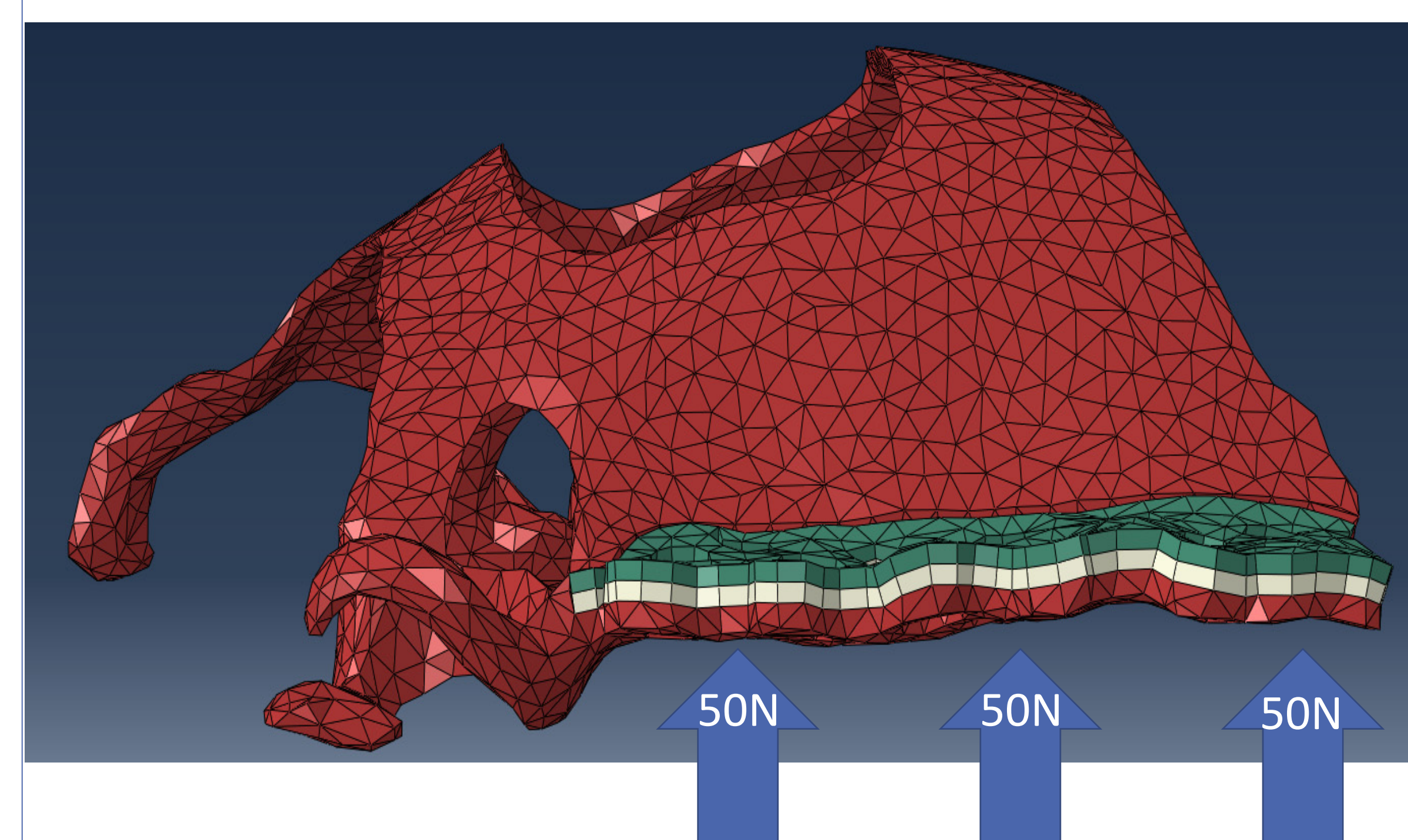
Materials

- 3D Scanned models: maxillary Class I (MxI), mandibular Class I (MnI), maxillary Class II (MxII), and mandibular Class II (MnII)
- SolidWorks (Dassault Systèmes), 3-matic (Materialise) and Abaqus (Dassault Systèmes)

Methods

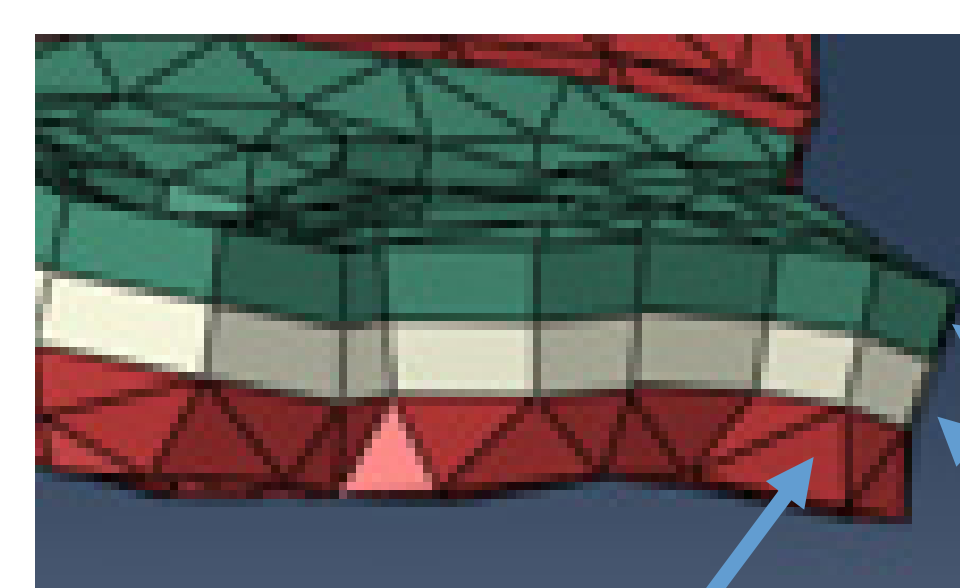


(Left) Model before in Solidworks, (Right) Model after in 3-matic



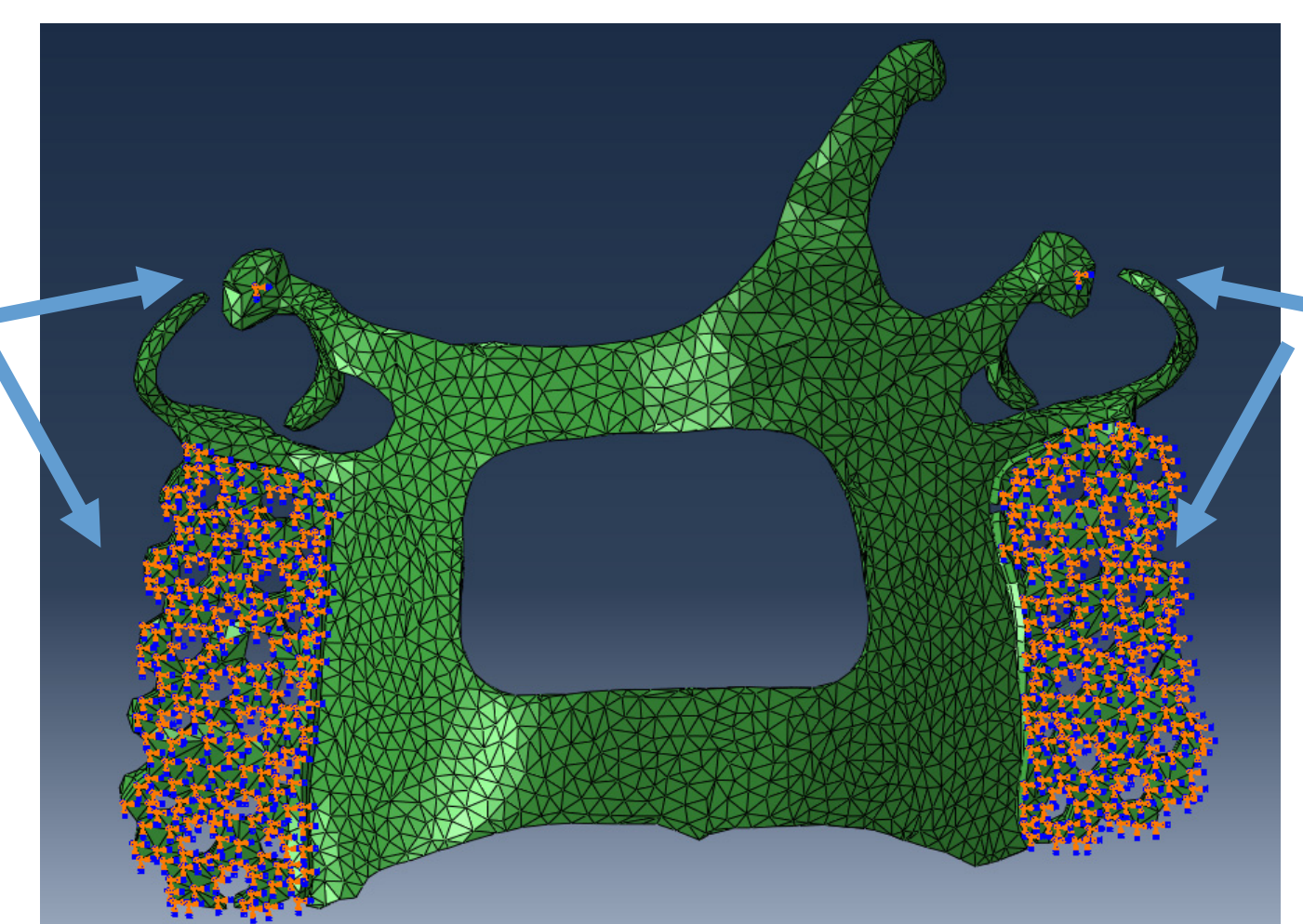
Load Applied: 50N to represent each tooth

Boundary Conditions:
Bone layer encastre, encastre at a single point on each direct retainer



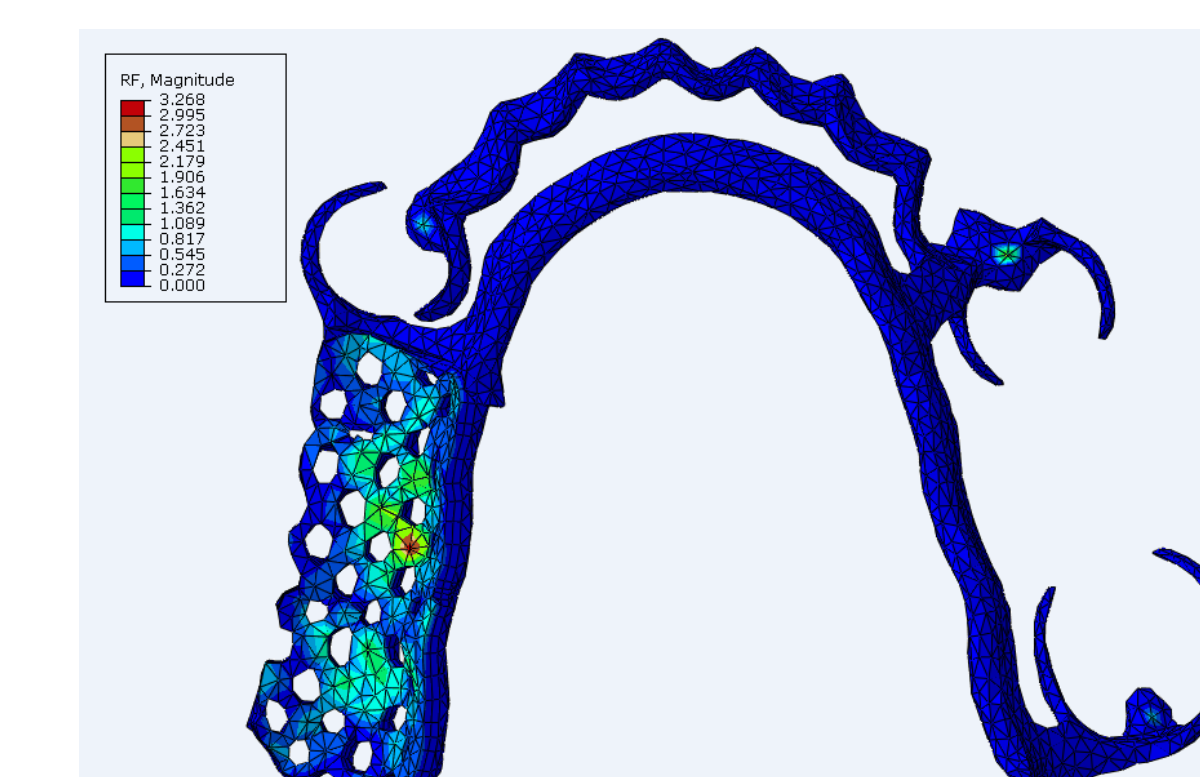
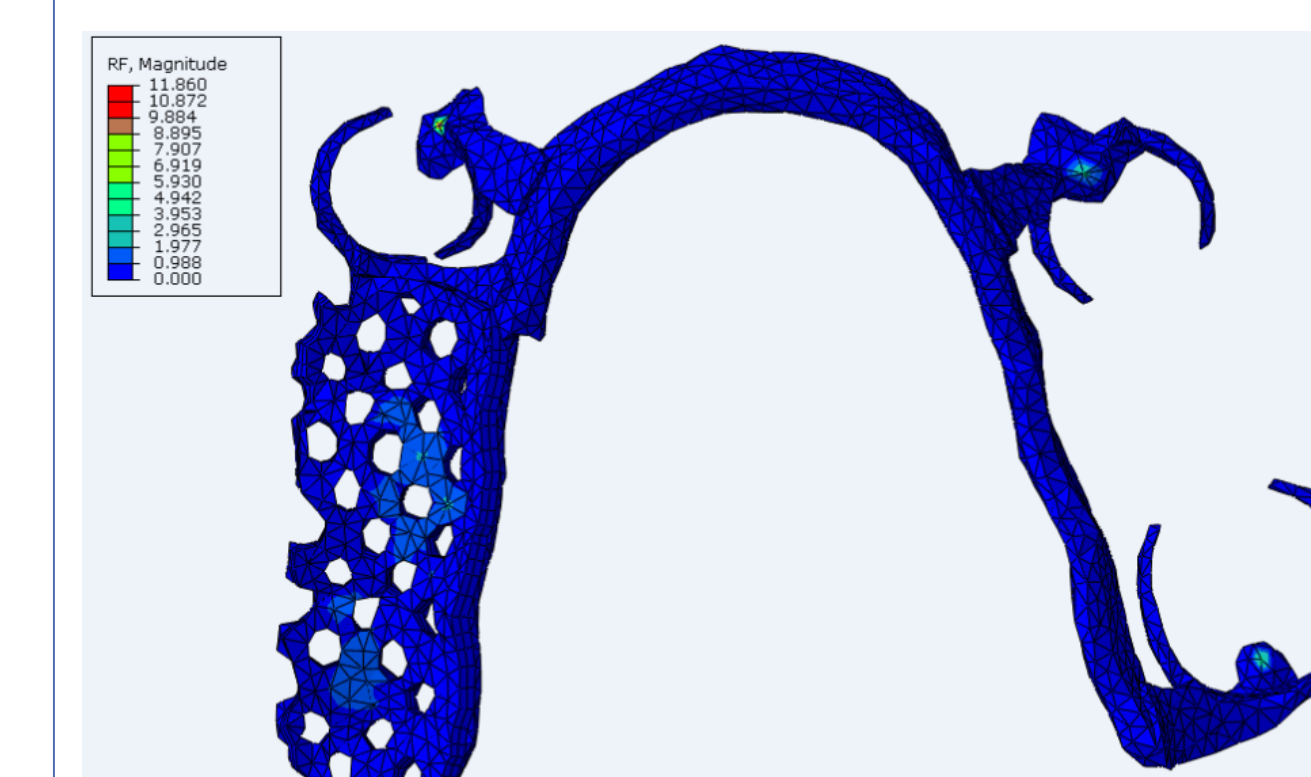
Vitallium (Dentsply)
(E=280 GPa, v=0.31)

Bone (E=100,000MPa, v=0.30)
Soft Tissue (E1=15MPa, E2=3MPa, v1=0.49, v2=0.35, Thickness1=0.5mm Thickness2=1.5mm)

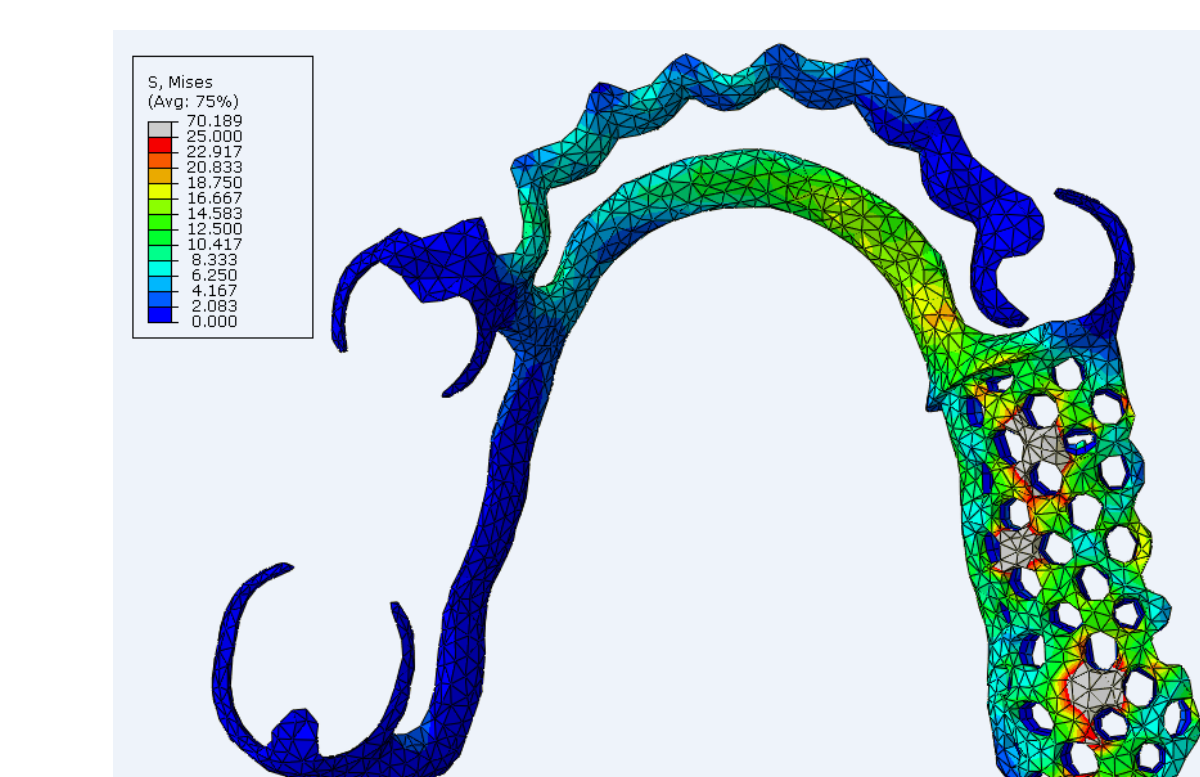
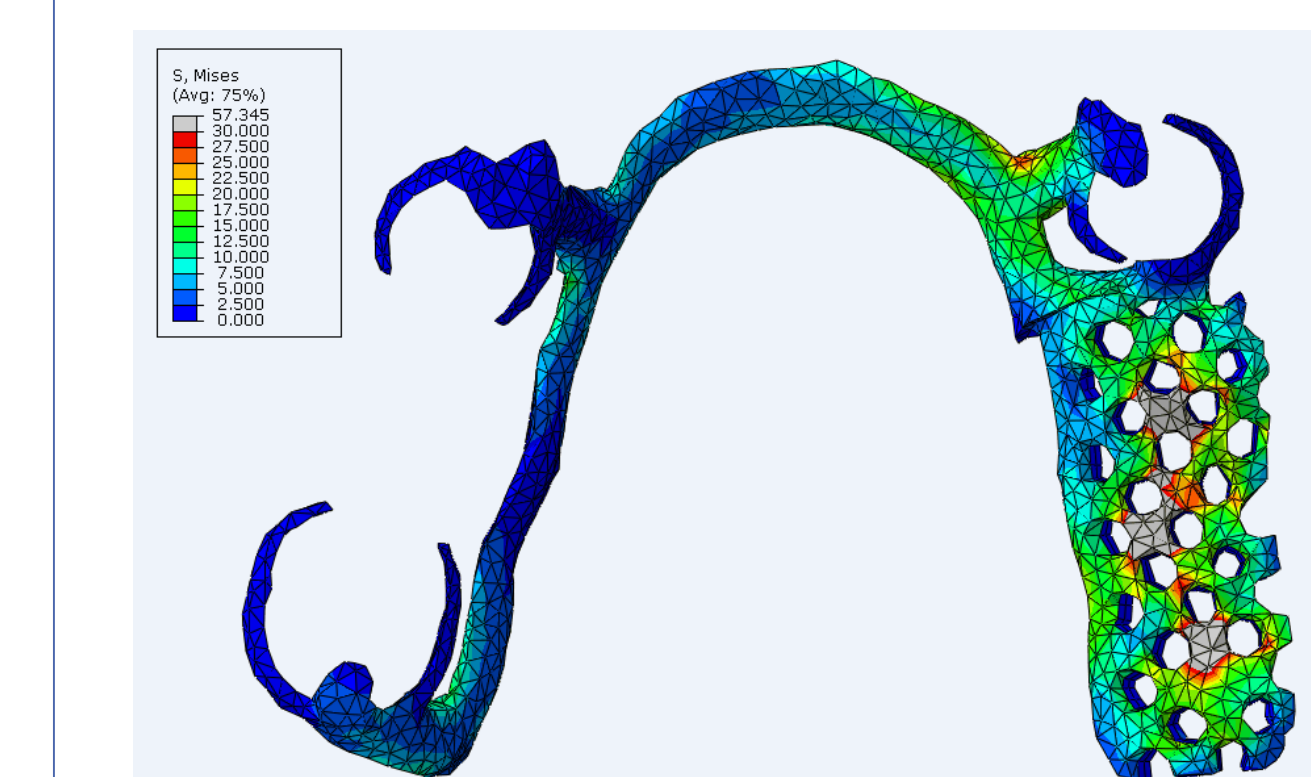


Results

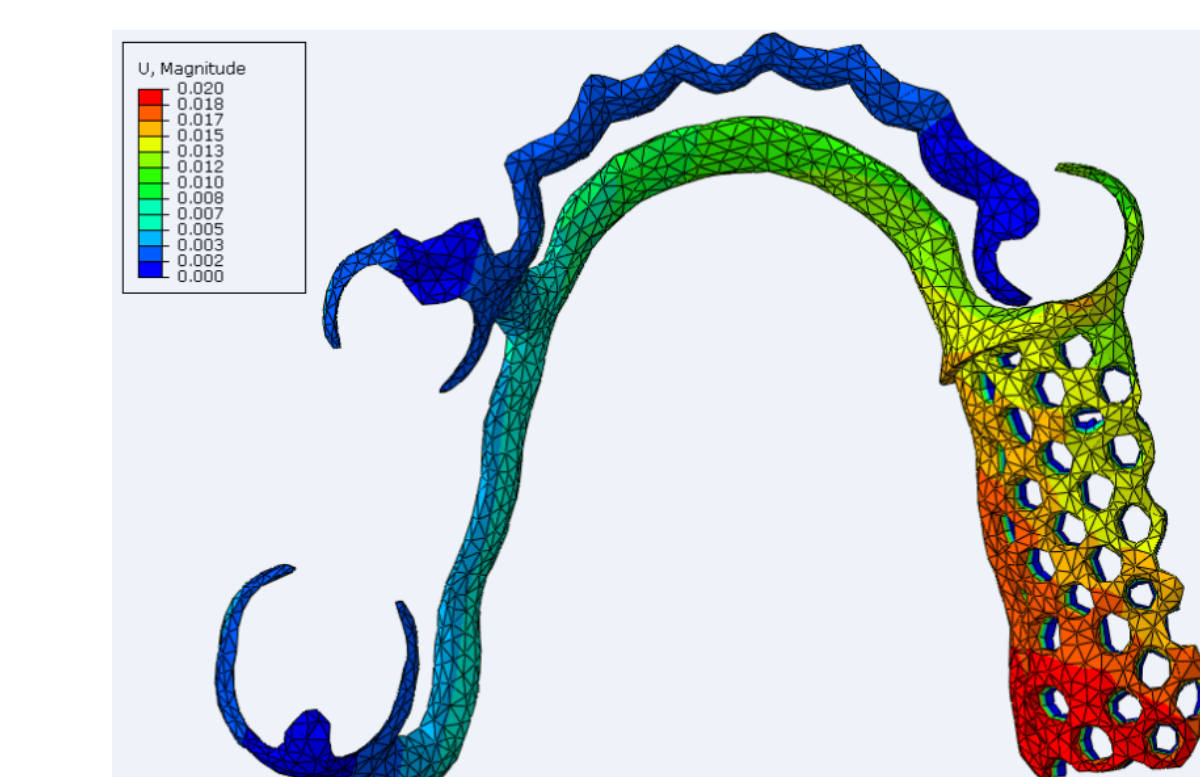
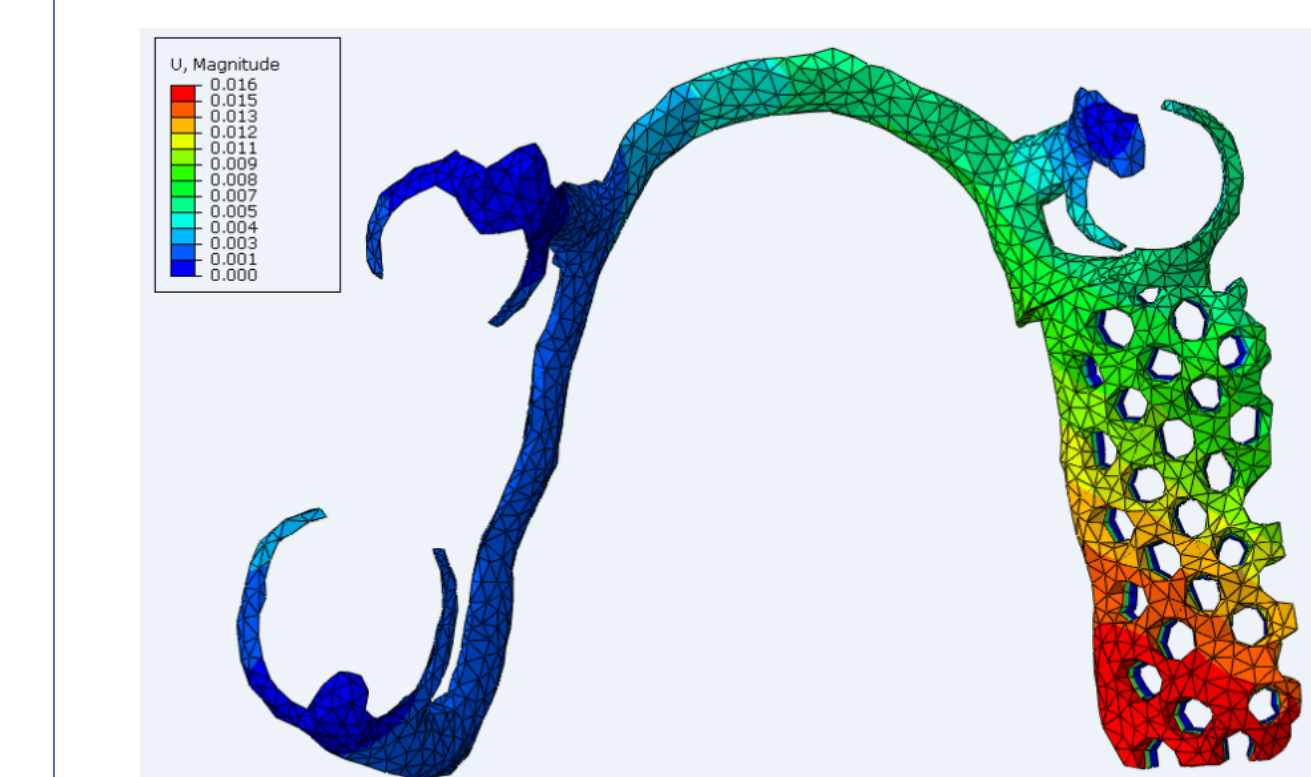
- Decrease in reaction forces in abutment teeth
- Greater tissue deformation and decreased stress/strain in MnI, MnII, and MxI.
- MxI changing design increased reaction forces



Reaction forces
(bottom view)



Stress
(top view)



Deformation
(top view)

Representative RPD FEAs: UL standard design (Left) and dual major connector design (right)

Discussion & Conclusion

- FEA is a powerful tool which can be used to compare various dental prosthetics
- Standardization can be maintained
- For MxI, design can be optimized using FEA to achieve desired decreased reaction forces

References

Chen, S. BMC Musculoskelet Disord (2013) 14: 281. Kanbara, R. Dent. Mater J (2013) 31.2: 297-308.